

Task 2:- Array Manipulation

Given an array $arr[]$, find the prefix sum of the same size, such that $prefix_sum[A]$ is $arr[0] + arr[1] + arr[2] + \dots + arr[i]$

~~explanations:-~~ for each

Sample input - 1: 5

10 20 10 5 15

Sample output - 1: 5

10 30 40 45 60

explanations:- for each index i , add all the elements from 0 to i .

Aims- To design and ~~imple~~^{imp}lement a java program that takes any array of integers as input and computes -

its Prefix sum Array where:

$$Prefix_sum[i] = \sum_{j=0}^i arr[j]$$

Algorithm:-

Step-1:- Start

Step-2:- Input the size of the array n

Step-3:- Create an array arr of size n and take n integers as input

Step-4:- Create a new array $prefix_sum$ of the same size.

Step-5:- set the first element: $prefix_sum[0] = arr[0]$

Step-6:- loop through the array from index $i=1$ to n
→ Calculate $prefix_sum[i] = prefix_sum[i-1] + arr[i]$

Step-7:- Find the elements of the prefix sum array

Step-8:- End

Input Enter the array size: 5

Enter array elements.

10 20 10 5 15

output

prefix sum array

0 20 30 55 50

Program:-

```
import java.util.Scanner;  
public class void prefix sum {  
    public static void main (String[] args) {  
        Scanner sc = new Scanner (System.in);  
        System.out.print ("Enter size of array:");  
        int n = sc.nextInt ();  
        int[] arr = new int [n];  
        int[] prefix sum = new int [n];  
        System.out.print ln ("Enter - array elements:");  
        for (int i=0; i<n; i++) {  
            arr[i] = sc.nextInt ();  
        }  
        prefix sum [i] = prefix sum [i-1] + arr[i] + arr[i]  
        prefix sum [0] = arr[0]  
        for (int i=1; i<n; i++) {  
            prefix sum [i] = prefix sum [i-1] + arr[i];  
        }  
        System.out.print ln ("Prefix sum Array:");  
        for (int x: prefix sum) {  
            System.out.print (x + " ");  
        }  
        sc.close();  
    }  
}
```

VEL TECH-CSE	
EX NO.	2
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	5
TOTAL (20)	20
SIGN WITH DATE	

Result:- Thus, the program is executed successfully.